

Paper 3 of 3 · Water pollution & agriculture

If Agriculture Paid

Food Prices, Export Competitiveness, and the Adequacy of Canada's Water-Pollution Cost Bucket

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Executive summary

Paper 1 estimated that Canadian agriculture externalizes approximately \$1.11 billion per year of the taxpayer pool for water-pollution management while paying \$22M (0.9%) of a \$2.57B public remediation ledger. That externalization (E_A) sits beside a separate private soil-productivity account (\$3.0B/yr) and a third public account: agri-environmental assistance to producers (\$0.9–1.0B/yr working, including Sustainable CAP / SCAP, OFCAF, GROW, Prime-Vert, and ASI), which is never added to the remediation ledger and never netted against a payor-transfer scenario. Status quo already includes partial provincial nutrient control (AOPA, BC AEM Code, PEI ACRA, Ontario NMA, Québec AOR) and sector-style financial assurance in mining, oil and gas, landfills, EPEA waste facilities, and digesters on the off-farm waste pathway. This paper asks what happens if remediation incidence shifts further onto the farm sector: how much retail food prices rise, how export margins move, and whether \$2.57B is an adequate social-damage account. A bounding cost-benefit model combines Paper 1–2 instrument rates with Canadian food pass-through evidence. Under the lead pass-through identity (3), a pure \$1.11B transfer raises retail food prices by about 0.24–0.79% across $\theta \in \{0.3, 0.5, 1.0\}$ (roughly \$30–\$100 per household per year at a \$12,500 food-spend scale), remaining a modest share of farm cash receipts (1.2%) and agri-food export value (1.1%). In South Saskatchewan irrigated corridors, administrative irrigation fees of about \$0.004–0.012/m³ sit beside return flows that carry nutrients into Bow and Oldman reaches, an attachment point for load-linked instruments (Government of Alberta 2014). Against documented eutrophication alone (\$0.36B), management spend exceeds that line ($A_{\text{doc}} \approx 7$); against upper social-damage working bands (\$4.5B), the bucket is incomplete. A transfer that leaves field loads unchanged reallocates remediation incidence while residual damage persists. Welfare ranking favors instruments that buy kilograms of source control incremental to practice change already funded by assistance, braided with existing provincial control and assurance templates from waste facilities and mining.

1. Introduction

Paper 1 in this series showed that, under central basin attribution, agriculture is assigned roughly 75% of nutrient and pesticide loading in Canada's major problem watersheds while paying about 0.9% of direct water-pollution management cost, a \$1.11 billion net externalization of the taxpayer pool from a \$2.57 billion public remediation ledger (Crandlemire 2026a). Three fiscal accounts must stay distinct from the outset (Table 1):

1. **Remediation ledger** (C_{total}): \$2.57B/yr public and ratepayer spend on treatment, monitoring, restoration, and documented eutrophication lines. Agriculture's attributed shortfall under 75% attribution is about \$1.91B; net externalization of the taxpayer pool alone is $E_A \approx \$1.11\text{B/yr}$ after \$22M remediation receipts.
2. **Private soil account**: \$3.0B/yr on-farm productivity loss from erosion and related degradation (Agriculture and Agri-Food Canada 2023). Reported beside the ledger for context; never summed with \$2.57B as a single welfare total.
3. **Agri-environmental assistance**: \$0.9–1.0B/yr working annualization of Sustainable CAP / SCAP, OFCAF, Manitoba GROW, Québec Prime-Vert, Ontario ASI, and related cost-share or ecological-goods payments (Agriculture and Agri-Food Canada, n.d.-a; 2025; -b; Government of Manitoba, n.d.; Gouvernement du Québec 2023; Ontario Soil and Crop Improvement Association, n.d.). A third **public** account: never added to C_{total} , never netted against E_A , and never subtracted from a transfer scenario. Transfer moves remediation incidence only.

Status quo already layers partial provincial nutrient control (AOPA / NRCB confined-feeding standards; BC AEM Code; PEI ACRA; Ontario NMA / CWA; Québec AOR) onto that assistance stack (Government of Alberta 2000; Government of Alberta 2014; Government of British Columbia 2019; Government of Prince Edward Island, n.d.; Government of Ontario 2002; 2006; Gouvernement du Québec, Ministère de l'Environnement 2025). Mining, oil and gas, landfills, and EPEA waste facilities post financial assurance before residual risks transfer to the public; anaerobic digesters follow the same logic along a manure-dominant AOPA pathway versus an EPEA waste-facility pathway when off-farm organics dominate (Bond Connect Canada, n.d.; Environmental Law Centre (Alberta), n.d.; Government of Alberta, n.d.-a; -b; Natural Resources Conservation Board 2023; Government of Alberta, Agriculture and Forestry, n.d.). Fee and transfer scenarios in this paper layer onto that baseline; they do not invent regulation from a blank field.

Paper 2 showed how portfolio-scale farmland ownership concentrates regulatory and litigation exposure when nutrient instruments arrive (Crandlemire 2026b). The open policy question is economic incidence beyond the farm gate: if agriculture paid a larger share of the remediation ledger, how would food prices and export competitiveness move, and is the current management budget large enough relative to social damage?

This paper addresses three linked questions with a bounding cost-benefit framework:

1. **Food prices**. How large is the retail food-price change when a \$1.11B/yr transfer (or a phosphorus-fee / nutrient-management package sized to Paper 2) lands on farm costs?
2. **Export competitiveness**. How large is the cost shock as a share of farm cash receipts and export unit value relative to peer exporters' nutrient policy, and how do irrigated return-flow corridors price water relative to nutrient load?
3. **Bucket adequacy**. Does \$2.57B in annual management expenditure cover plausible social-damage bands, or does transfer alone leave residual damage largely intact?

Contribution. The paper (i) links Paper 1–2 dollar instruments to a transparent food pass-through identity grounded in Canadian evidence (Tombe and Winter 2024; Hyun and Lee 2023; Chen et al. 2025); (ii) bounds export-margin effects using cash-receipt and agri-food export comparators, with irrigation administrative fees and Bow–Oldman return-flow attachment for load-linked instruments (Rivers and Schaufele 2013; Agriculture and Agri-Food Canada and Statistics Canada 2024; Government of Alberta 2014); (iii) defines a damage-adequacy ratio against Dodds-scaled and omitted-category damage bands (Dodds et al. 2009); and (iv) separates **transfer** (who pays today’s remediation ledger) from **abatement** (shrinking residual damage incremental to assistance-funded practice change), showing that welfare ranking concentrates on the second path (Oleszkiewicz and others 2015; Garcia-Hernandez et al. 2022).

Section 2 reviews pass-through, competitiveness, and the fiscal and regulatory baseline. Section 3 states the model. Section 4 reports results. Section 5 discusses policy sequencing. Section 6 lists limitations. Section 7 concludes.

2. Background and Related Work

2.1 Three fiscal accounts and Papers 1–2 instruments

Figure 1 restates the Paper 1 incidence story as paired bars: attributed pollution cost (sector share $\times$ \$2.57B ledger) versus cost covered. Agriculture is attributed \$1,929M and pays \$22M; municipal and industrial payors cover above their shares; taxpayers fund \$1,506M with 0% attributed load, carrying \$1.11B net externalization of the taxpayer pool (Crandlemire 2026a). Table 1 keeps the remediation ledger, private soil account, and agri-environmental assistance on separate books. Assistance annualization (\$0.9–1.0B/yr working) exceeds agriculture’s \$22M remediation receipts by more than an order of magnitude; both dollars are public, and only the \$22M sits inside C_{total} .

Table 1: Three fiscal accounts carried from Paper 1 into this paper’s scenarios. Assistance is never added to C_{total} and never netted against transfer.

Account	Central / working scale	Role in Paper 3
Remediation ledger (C_{total})	\$2.57B/yr; $C_A = \$22\text{M}$; $E_A \approx \$1.11\text{B}$	Transfer base; incidence only
Private soil productivity	\$3.0B/yr (Agriculture and Agri-Food Canada 2023)	Reported beside ledger; outside welfare add
Agri-environmental assistance	\$0.9–1.0B/yr working (Agriculture and Agri-Food Canada, n.d.-a; 2025; Government of Manitoba, n.d.; Gouvernement du Québec 2023; Ontario Soil and Crop Improvement Association, n.d.)	Third public account; net C_{abate} against funded BMPs

Paper 2 supplies unit instruments for a 100,000-acre reference holding: phosphorus fee \$15/kg on 4 kg/ha surplus ($F_P \approx \$2.43\text{M/yr}$) and nutrient-management compliance ($C_{\text{NMP}} \approx \$1.73\text{M/yr}$) (Crandlemire 2026b). This paper nationalizes the dollar transfer first (the \$1.11B taxpayer-pool externalization), then tests fee/NMP sensitivity at Paper 2 rates.

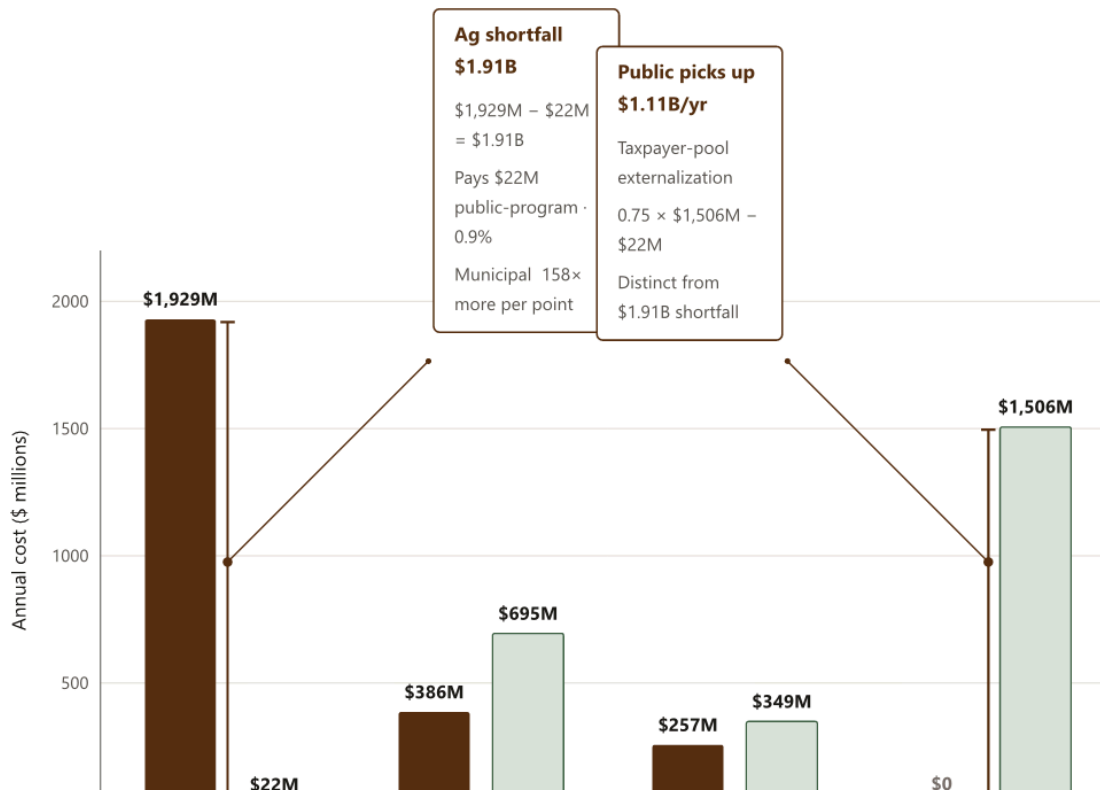


Figure 1: Attributed pollution cost versus cost covered by sector under Paper 1 central attribution (75 / 15 / 10% of the \$2.57B ledger). Agriculture's shortfall and the \$1.11B taxpayer-pool externalization are the transfer base for this paper's scenarios.

2.2 Existing nutrient control and assurance peers

Provincial nutrient rules already bind defined operation classes. Québec's **Regulation respecting agricultural operations** (AOR) requires agro-environmental fertilization plans and annual phosphorus reports for specified livestock and spreading operations (Gouvernement du Québec, Ministère de l'Environnement 2025). Ontario's **Nutrient Management Act, 2002** (NMA) and **Clean Water Act, 2006** (CWA) source-protection plans impose certified nutrient management strategies and plans on larger livestock operations and drinking-water-threat activities (Government of Ontario 2002; 2006). Alberta's **Agricultural Operation Practices Act** (AOPA) sets manure storage, setback, incorporation, soil nitrate/salinity, and land-base standards for confined feeding operations, administered by the Natural Resources Conservation Board (NRCB), and supports middle-reach phosphorus management where irrigation return flows, confined feeding, and municipal wastewater share Bow and Oldman corridors (Government of Alberta 2000; -c; 2014). British Columbia's **Code of Practice for Agricultural Environmental Management** (AEM Code) applies to all agricultural operations regardless of size (Government of British Columbia 2019). Prince Edward Island's **Agricultural Crop Rotation Act** (ACRA) requires three-year rotations for potatoes and row crops, with buffer and slope limits (Government of Prince Edward Island, n.d.). Thresholds and cropland coverage differ by province; fee, discharge-linked, and transfer instruments in this paper layer onto that partial control baseline.

Mining requires reclamation bonds covering estimated closure costs (Bond Connect Canada, n.d.). Oil and gas operators post abandonment security (Environmental Law Centre (Alberta), n.d.). Landfills and commercial waste facilities under Alberta's **Environmental Protection and Enhancement Act** (EPEA) must post financial security for closure and reclamation before operations begin (Government of Alberta, n.d.-a; -b). Anaerobic

digesters follow the same assurance logic along an on-farm / off-farm line: manure-dominant digesters (at least 50% manure by wet weight, with listed co-feedstocks) can store and land-apply digestate under AOPA/NRCB rules, while plants that take substantial off-farm organics are typically approved as EPEA waste-management facilities and carry Director-set reclamation security (Natural Resources Conservation Board 2023; Government of Alberta, Agriculture and Forestry, n.d.; Government of Alberta 2000). Farm-level bonding calibrated to diffuse water-quality outcomes at the outlet remains outside that peer set; the peer template for remediation finance and posted security is the waste-facility and mining model.

2.3 Food-price pass-through

Farm cost shocks reach households through a long processing and retail chain. Tombe and Winter quantify Canadian emissions pricing effects on food production costs and retail prices: full pricing of agricultural emissions can raise sector costs several percent, while current exemption design cuts average food effects to roughly 0.5–1.1% (Tombe and Winter 2024). Chen et al. show that high-value-added food systems mute the link from farm-gate prices to retail food (Chen et al. 2025). Bank of Canada staff estimate long-run pass-through from global agricultural commodity prices to domestic food inflation near 0.12–0.15 per 1% commodity move, with higher coefficients for net exporters (Hyun and Lee 2023). Those sources motivate a bounded pass-through parameter θ between farm cost and shelf price. The Hyun elasticity is a commodity-price % $\rightarrow$ food-inflation % relationship; here θ is used as an attenuation factor on a dollar shock normalized by the retail food bill (Form A) or by farm receipts with an explicit farm-share lower band (Form B).

2.4 Export competitiveness and water pricing

Competitiveness objections often treat environmental cost as a unique trade threat. Rivers and Schaufele find little evidence that British Columbia's carbon tax produced large negative agricultural export effects, which weakens the case for blanket exemptions justified solely on trade grounds (Rivers and Schaufele 2013). Peer exporters already price nutrients more tightly through EU CAP conditionality, New Zealand freshwater nutrient limits, and US state nutrient programs; Papers 1 and 2 survey those precedents. This paper compares cost-shock percentages under Canadian scenarios against farm receipts and agri-food export value.

Water-pricing context matters for irrigated export crops. In South Saskatchewan corridors, irrigation water is priced at approximately \$0.004–0.012 per cubic metre as administrative fees under irrigation-district statutes, even where return flows convey nutrients into phosphorus-stressed Bow and Oldman reaches (Government of Alberta 2014). That volumetric band is low relative to municipal and industrial true-cost benchmarks in Paper 1; the policy attachment for load-linked instruments is the return-flow and farm-outlet pathway those fees already accompany, rather than fee height alone.

2.5 Management spend versus social damage

Paper 1's eutrophication line (\$362M) is a documented, conservative damage component inside the \$2.57B ledger; Dodds et al. estimated US freshwater eutrophication damages near \$2.2B/yr (Dodds et al. 2009). Canadian basins hold a disproportionate share of severe large-lake eutrophication, so population-scaled and freshwater-area-adjusted brackets sit above Paper 1's bottom-up eutrophication build. Groundwater, fisheries, and Indigenous harms remain largely outside the monetized accounts (Crandlemire 2026a). Adequacy, defined below, compares management expenditure to those damage bands.

3. Methods

3.1 Instrument scenarios

Five policy cases structure the results (Table 2):

1. **Status quo.** Agriculture remits \$22M remediation receipts inside the \$2.57B ledger; operates under partial provincial nutrient control (AOPA, BC AEM Code, PEI ACRA, Ontario NMA / CWA, Québec AOR); and receives agri-environmental assistance (\$0.9–1.0B/yr working) as a separate public account. Load and payor shares follow Paper 1.
2. **Transfer.** Agriculture remits an additional \$1.11B/yr equal to Paper 1 net externalization of the taxpayer pool (central 75% attribution). Assistance dollars and private soil costs stay on their own accounts; the transfer moves remediation incidence only.
3. **Fee.** National phosphorus fee at Paper 2 base \$15/kg, with \$5–\$50/kg sensitivity; national load scaled so fee revenue is comparable to the transfer band (circular by design; used only as instrument-shape sensitivity). Layers onto existing provincial control.
4. **Fee + NMP.** Adds Paper 2 nutrient-management compliance intensity, scaled from the 100,000-acre reference.
5. **Transfer + abatement.** Transfer paired with illustrative 10% or 25% field-load reduction at CWN-style abatement costs, counted as incremental net of practice change already funded by assistance.

Results lead with the transfer scenario because it is dollar-identical to the Paper 1 inequity claim; fee, NMP, and abatement provide instrument-design and welfare ranking sensitivity.

Table 2: Baseline and policy scenarios (lead case dollars).

Scenario	Farm cost shock / fiscal position	Load change
Status quo	\$22M remediation receipts + partial provincial control + assistance received	Unchanged (Paper 1 load)
Transfer	+\$1.11B/yr remediation remittance	Unchanged
Fee (\$15/kg band)	Revenue scaled $\approx$ transfer; layers on control	Behavioral response possible
Fee + NMP	Fee + scaled NMP	Compliance-driven
Transfer + 10% / 25% abate	+\$1.11B + incremental C_{abate}	–10% / –25% field load

3.2 Farm cost shock

$$\Delta C_{\text{farm}} = T + F_P^{\text{nat}} + C_{\text{NMP}}^{\text{nat}}, \quad (1)$$

where T is the transfer scenario (\$1.11B in the lead case), and the fee and NMP terms are zero or positive depending on scenario. Assistance is outside (1). Share of farm cash receipts:

$$s_{\text{receipts}} = \Delta C_{\text{farm}} / R_{\text{farm}}, \quad (2)$$

with central $R_{\text{farm}} = \$90\text{B}$ (Statistics Canada farm cash receipts working scale (Statistics Canada 2024a); pin latest table ID in the workbook).

3.3 Food-price pass-through

Lead identity (Form A). A dollar farm-cost shock normalized by the retail food bill, attenuated by θ :

$$\Delta P_{\text{food}}^A (\%) = 100 \times \theta \times (\Delta C_{\text{farm}} / R_{\text{food}}). \quad (3)$$

Lower band (Form B). If only the farm-gate share of the consumer food dollar transmits the shock into domestic groceries:

$$\Delta P_{\text{food}}^B (\%) = 100 \times \theta \times s_{\text{farm}} \times (\Delta C_{\text{farm}} / R_{\text{farm}}), \quad (4)$$

where s_{farm} is the farm-gate share of the consumer food dollar (central 0.12; band 0.08–0.18), R_{food} is annual retail food expenditure (central \$140B working scale (Statistics Canada 2024b)), and θ is pass-through from farm cost to retail food (central 0.5; band 0.3–1.0), informed by (Tombe and Winter 2024; Hyun and Lee 2023; Chen et al. 2025). Form A is an upper-bound domestic CPI path if the full \$1.11B lands in the food chain; Form B is a lower band when the shock is scaled by farm share of receipts. Household cost:

$$\Delta H = (\Delta P_{\text{food}} / 100) \times E_{\text{hh}}, \quad (5)$$

with central household food expenditure $E_{\text{hh}} = \$12,500/\text{yr}$ (workbook-editable).

3.4 Export unit-cost bound

$$s_{\text{export}} = \Delta C_{\text{farm}} / V_{\text{export}}, \quad (6)$$

where $V_{\text{export}} = \$99\text{B}$ is annual Canadian agri-food export value (working scale (Agriculture and Agri-Food Canada and Statistics Canada 2024); canola, wheat, beef, pork, and soy dominate). The competitiveness test is whether s_{export} and s_{receipts} sit inside normal input-cost noise (Rivers and Schaufele 2013). For irrigated export production in South Saskatchewan corridors, administrative water fees of about $\$0.004\text{--}0.012/\text{m}^3$ provide the conveyance pricing baseline against which nutrient-linked instruments attach at the return-flow pathway (Government of Alberta 2014).

3.5 Damage adequacy

$$A = C_{\text{mgmt}} / D, \quad (7)$$

with $C_{\text{mgmt}} = \$2.57\text{B}$ from Paper 1, D_{doc} = documented eutrophication $\$0.36\text{B}$, $D_{\text{low}} = \$1.0\text{B}$ and $D_{\text{high}} = \$4.5\text{B}$ as working social-damage bands (Dodds-scaled Canadian freshwater damages plus omitted categories; bands are working assumptions, pending audited national damage accounts). $A_{\text{doc}} = 2.57/0.36 \approx 7.1$: management spend exceeds the documented eutrophication line. $A_{\text{high}} = 2.57/4.5 \approx 0.57$: against the upper band the bucket is incomplete. Adequacy uses the remediation ledger only; assistance and private soil stay off A .

3.6 Welfare identity

$$\Delta W = -C_{\text{abate}} + \Delta D_{\text{avoided}} - \Delta \text{CS}_{\text{food}} + G_{\text{fiscal}}, \quad (8)$$

where C_{abate} is **incremental** field abatement cost net of practice change already funded by agri-environmental assistance, $\Delta D_{\text{avoided}}$ is reduced residual damage, $\Delta \text{CS}_{\text{food}}$ is consumer surplus loss from food-price rise, and G_{fiscal} is the fiscal gain to taxpayers/municipalities from reduced remediation externalization. Pure transfer

sets $C_{\text{abate}} = 0$ and $\Delta D_{\text{avoided}} \approx 0$; after consumer-surplus loss, a pure remittance is largely redistribution of remediation incidence. Efficiency gains concentrate where incremental kilograms of load leave the field. Figure 4 therefore reports ranking only (index = 1 for transfer-only); absolute \$B welfare levels remain scenario products. Abatement cases rank higher where source control is cheaper per kilogram than end-of-pipe polishing (Oleszkiewicz and others 2015; Garcia-Hernandez et al. 2022).

Incidence note. Fees and transfers may capitalize into land rents: landlords capture part of the shock under cash rent, tenants under crop share, consumers via Form A/B food prices. The lead results do not split those shares; they bound the consumer and farm-receipt ends.

4. Results

4.1 Food prices under a \$1.11B transfer

Applying (3) (Form A) with $\Delta C_{\text{farm}} = \1.11B , $R_{\text{food}} = \$140\text{B}$, and θ in $\{0.3, 0.5, 1.0\}$ yields retail food-price changes of **0.24%**, **0.40%**, and **0.79%** (Figure 2; Table 3). At $E_{\text{hh}} = \$12,500$, that is about \$30, \$50, and \$99 per household per year. Form B (4) with $s_{\text{farm}} = 0.12$ and $R_{\text{farm}} = \$90\text{B}$ yields a lower band of **0.04%**, **0.07%**, and **0.15%** (about \$5–\$19/hh). The Form A central case (0.40%) sits below Tombe–Winter’s exemption-design food-effect band (0.5–1.1%) for a much smaller fiscal shock than full agricultural emissions pricing (Tombe and Winter 2024). Low-income households spend a larger share of budgets on food; Table 4 reports a crude quintile sketch using higher food-at-home shares at the bottom of the income distribution.

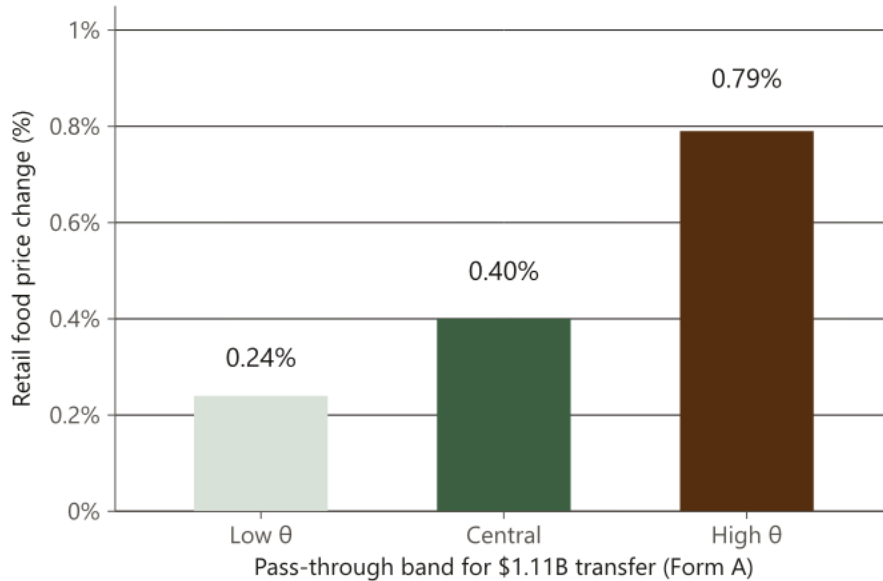


Figure 2: Retail food-price change (%) under a \$1.11B/yr farm cost transfer for low, central, and high pass-through θ (Form A, (3)).

Table 3: Food-price pass-through under a \$1.11B transfer (Form A lead; Form B lower band).

θ	ΔP Form A (%)	ΔH Form A (\$/yr)	ΔP Form B (%)	ΔH Form B (\$/yr)
0.3	0.24	30	0.04	5
0.5	0.40	50	0.07	9
1.0	0.79	99	0.15	19

Table 4: Illustrative household food-cost hit by income quintile (Form A, $\theta = 0.5$; food-spend shares stylized).

Quintile	Food spend (\$/yr)	ΔH (\$/yr)	Share of food spend (%)
Q1 (lowest)	8,000	32	0.40
Q3	12,500	50	0.40
Q5 (highest)	18,000	72	0.40

Fee scenarios that raise farm costs above \$1.11B scale approximately linearly in (3) until behavioral abatement reduces load. Tombe–Winter exemption results caution that instrument design (who is charged at the farm versus input manufacturer) can cut food effects by most of a full-pricing shock (Tombe and Winter 2024).

4.2 Export competitiveness

At $R_{\text{farm}} = \$90\text{B}$, $s_{\text{receipts}} = 1.11/90 \approx 1.2\%$ (2). At $V_{\text{export}} = \$99\text{B}$, $s_{\text{export}} = 1.11/99 \approx 1.1\%$ (6). Canola, wheat, beef, pork, and soy account for the bulk of that export value (Agriculture and Agri-Food Canada and Statistics Canada 2024). A 1.1–1.2% unit-cost shock is a visible margin hit for tight-cost grains and oilseeds operations, yet it sits inside the range of fertilizer and fuel cost swings operators already manage season to season. Relative to Paper 2’s reference holding, \$15/kg fees absorb a large share of BRM receipts at the parcel scale (Crandlemire 2026b); portfolio landlords and highly leveraged tenants feel the shock first. Rivers and Schaufele’s carbon-tax trade evidence suggests modest environmental cost shocks can clear through price with limited export-volume collapse (Rivers and Schaufele 2013).

Irrigated districts in the South Saskatchewan system add a second margin channel: administrative irrigation fees of about \$0.004–0.012/m³ convey water, and return flows convey nutrients, into Bow and Oldman middle reaches already under multi-source phosphorus management (Government of Alberta 2014). Export-unit cost bounds in (6) capture fee and transfer shocks on farm costs; load-linked instruments attach at the same return-flow pathway that cheap conveyance already serves.

4.3 Abatement cost order of magnitude

Table 5 summarizes published order-of-magnitude \$/kg phosphorus (or TP) removal costs. Field and watershed practices are typically cheaper than the last increments of municipal polishing (Oleszkiewicz and others 2015); Great Lakes basin modelling finds agricultural TP abatement cheaper at the margin than manufacturing point-source cuts for many targets (Garcia-Hernandez et al. 2022). Manitoba pig phosphorus-regulation compliance costs of roughly CAD \$18–28M/yr for that sector alone illustrate that private nutrient rules already impose material operator costs when they bind (Yap and others 2015). Incremental C_{abate} in (8) excludes BMP outlays already covered by SCAP, OFCAF, GROW, Prime-Vert, ASI, and similar assistance.

Table 5: Illustrative phosphorus abatement cost orders of magnitude (cited ranges; workbook scenario inputs).

Measure	Order of magnitude	Source
Field BMPs / watershed source control	Lower \$/kg TP than polishing	(Oleszkiewicz and others 2015)
Municipal tertiary polishing (last increments)	Higher \$/kg TP	(Oleszkiewicz and others 2015)
Great Lakes ag vs manufacturing TP cut	Ag MAC lower for many targets	(Garcia-Hernandez et al. 2022)
Manitoba pig P-compliance (sector)	CAD \$18–28M/yr	(Yap and others 2015)

4.4 Bucket adequacy

Figure 3 compares \$2.57B management spend to documented eutrophication (\$0.36B) and to \$1.0–\$4.5B social-damage working bands. Adequacy $A_{\text{doc}} \approx 7.1$ means spend exceeds the documented eutrophication line; $A_{\text{high}} \approx 0.57$ means the upper damage band exceeds the management bucket. Transferring who pays for \$2.57B changes remediation incidence; residual damage stays in place unless load falls. Assistance (\$0.9–1.0B/yr) and private soil (\$3.0B) remain outside A .

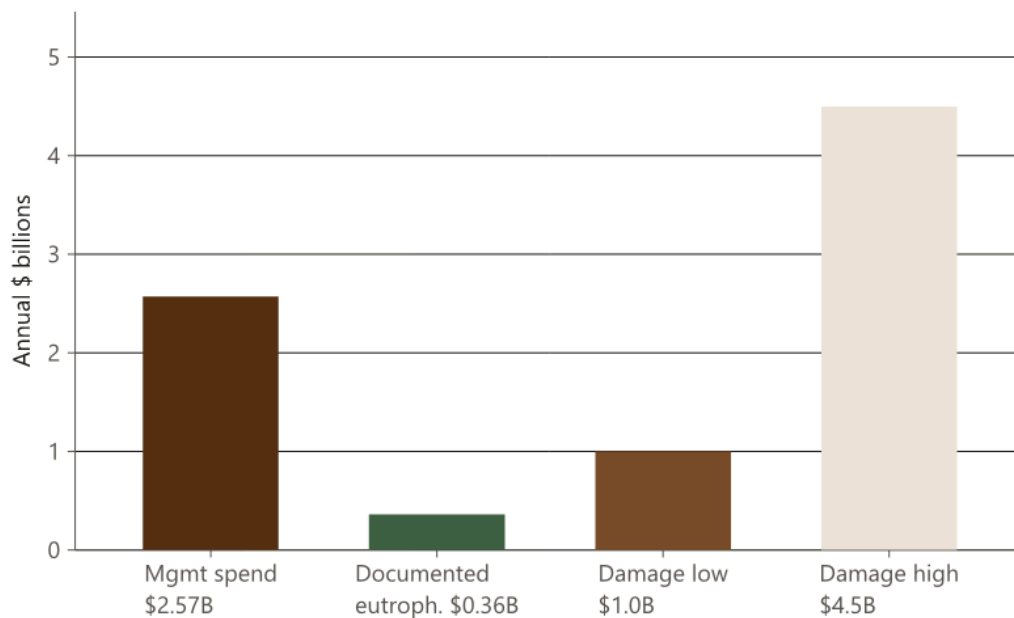


Figure 3: Damage-adequacy comparison: public management spend (\$2.57B) against documented eutrophication (\$0.36B) and low/high social-damage working bands (Z).

4.5 Transfer versus abatement

Figure 4 reports ranking under (8): transfer only (index = 1); transfer plus 10% load abatement; transfer plus 25% load abatement. Absolute \$B levels are omitted; the claim is ordinal. Ranking rests on CWN and Great Lakes evidence that field practices often remove nutrients at lower cost per kilogram than polishing the last increments at municipal plants (Oleszkiewicz and others 2015; Garcia-Hernandez et al. 2022). Sequencing implication: braid transfer instruments with measurable incremental abatement (buffers, cover crops, soil-

test ceilings, verified NMP) and with existing provincial control and assistance, so fiscal recycle and damage reduction move together.

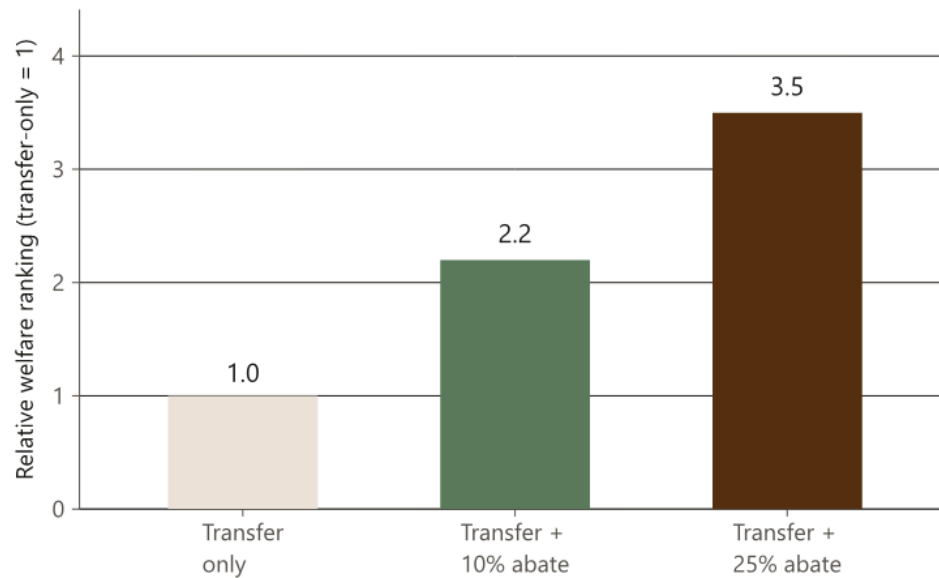


Figure 4: Relative welfare ranking (transfer-only = 1) under transfer versus transfer-plus-abatement scenarios (8). Ordinal claim only; absolute \$B welfare remains a scenario product.

5. Discussion

5.1 Answers to the three questions

Food cost. Under Form A central pass-through ($\theta = 0.5$), a \$1.11B agricultural remittance raises retail food prices by about 0.40% (\$50/hh/yr), enough to warrant equity attention for low-income households (Table 4), small relative to the Paper 1 taxpayer externalization and to long-run soil-driven food-cost paths (Soil Conservation Council of Canada 2016). Form B keeps the hit under 0.1% at central θ .

Export ability. At roughly 1.2% of farm cash receipts and 1.1% of agri-food export value, the transfer is a margin event, concentrated on intensive and portfolio operations highlighted in Paper 2. Historical Canadian evidence on environmental cost and trade cautions against assuming large export-volume collapse (Rivers and Schaufele 2013). In irrigated South Saskatchewan corridors, administrative fees of about \$0.004–0.012/m³ and nutrient-bearing return flows into Bow and Oldman reaches mark the hydrologic attachment for load-linked instruments alongside unit-cost bounds (Government of Alberta 2014).

Bucket size. The \$2.57B program exceeds documented eutrophication (\$0.36B) and sits below upper social-damage working bands. Adequacy therefore depends on the damage frontier chosen. Payor shifts that leave field loads unchanged leave the damage frontier intact. Private soil (\$3.0B) and agri-environmental assistance (\$0.9–1.0B/yr) stay on their own accounts.

5.2 Sequencing

Instruments that recycle the \$1.11B toward watershed source control raise both adequacy (lower residual damage) and welfare ranking (8), provided C_{abate} counts only incremental practice change beyond assistance already received. Fee revenue dedicated to end-of-pipe upgrades with unchanged field loads improves payor equity while leaving the damage frontier high, consistent with Paper 1's finding that public treatment is an expensive substitute for field control (Oleszkiewicz and others 2015; Crandlemire 2026a).

Braid sequence: (i) keep provincial nutrient control (AOPA, AEM Code, ACRA, NMA, AOR) as the practice baseline; (ii) keep SCAP / OFCAF / GROW / Prime-Vert / ASI as the funded-BMP account, separate from remediation remittance; (iii) layer transfer or fee instruments onto remediation incidence only; (iv) borrow assurance design from landfills, EPEA waste facilities, mining reclamation bonds, and digesters on the off-farm EPEA pathway where residual environmental liability is posted before public absorption (Government of Alberta, n.d.-a; -b; Bond Connect Canada, n.d.; Natural Resources Conservation Board 2023).

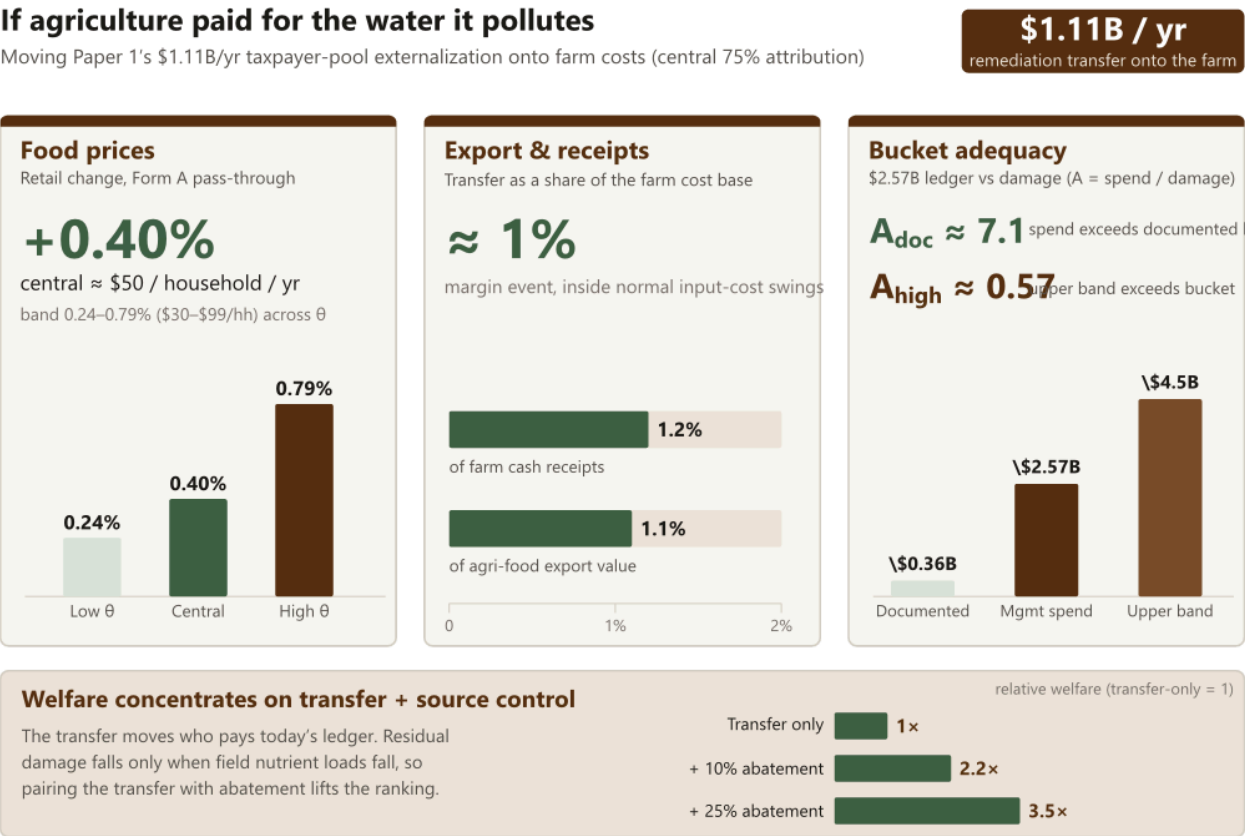
Co-benefit rule. Greenhouse-gas and soil-carbon co-benefits of nutrient BMPs are excluded from the lead welfare ranking and may be added only as a sensitivity case with an explicit \$/t CO₂e assumption in the workbook.

6. Limitations

The model is bounding arithmetic under partial-equilibrium incidence assumptions. National phosphorus surplus inventories suitable for exact fee nationalization are incomplete. R_{farm} , R_{food} , and V_{export} are working scales pinned in the accompanying workbook against Statistics Canada releases (Statistics Canada 2024a; 2024b; Agriculture and Agri-Food Canada and Statistics Canada 2024). Indigenous, groundwater, and cultural water harms remain qualitative. Figure 4 communicates ranking under CWN-style abatement costs; absolute \$B welfare remains a scenario product. Competitiveness comparisons are unit-cost and peer-policy bounds rather than gravity-model trade elasticities. Land-rent capitalization between landlords and tenants is noted and left unquantified. Assistance annualization (\$0.9–1.0B/yr) is a working bound from multi-year envelopes, pending an audited national line item.

7. Conclusion

If Canadian agriculture remitted the \$1.11B now externalized onto the taxpayer pool, retail food prices would rise by tenths of a percent under Form A Canadian pass-through bounds (and less under Form B), export margins would tighten within the range of ordinary input-cost volatility for many commodities, and the \$2.57B management ledger would remain an incomplete social-damage account against upper damage bands until field-level abatement shrinks residual harm. Transfer moves remediation incidence only; agri-environmental assistance and private soil stay on their own accounts; existing provincial control and waste-facility / mining assurance templates supply the institutional baseline. The cost-benefit case for polluter-pays in agriculture is transfer plus incremental source control. Papers 1 and 2 established who pays and who inherits enforcement risk; this paper shows that making agriculture pay is compatible with manageable food and trade effects when instruments buy kilograms of reduced load.



10. Abbreviations

Introduced abbreviations used more than once. Full titles appear at first use in the body.

Table 6: Abbreviations used in this paper.

Abbreviation	Expansion
AAFC	Agriculture and Agri-Food Canada
ACRA	Agricultural Crop Rotation Act (Prince Edward Island)
AEM	Agricultural Environmental Management (British Columbia Code of Practice)
AOPA	Agricultural Operation Practices Act (Alberta)
AOR	Regulation respecting agricultural operations (Québec)
ASI	Agricultural Stewardship Initiative (Ontario)
BMP	Beneficial management practice
BRM	Business risk management
CAP	Common Agricultural Policy (European Union)
CWA	Clean Water Act, 2006 (Ontario)
CWN	Canadian Water Network
EPEA	Environmental Protection and Enhancement Act (Alberta)
GROW	Growing Outcomes in Watersheds (Manitoba)
MAC	Marginal abatement cost
NMA	Nutrient Management Act, 2002 (Ontario)
NMP	Nutrient management plan
NRCB	Natural Resources Conservation Board (Alberta)
OFCAF	On-Farm Climate Action Fund
PMP	Phosphorus Management Plan (Bow River)
SCAP / Sustainable CAP	Sustainable Canadian Agricultural Partnership (2023–2028)
TP	Total phosphorus

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